

Mark Scheme — Practice Paper 6 (P1)

M = method mark, A = accuracy mark, B = independent mark. 教师/批改用。

1 $(x+5)(x-3) > 0$ (M1, A1 critical values). $x < -5$ or $x > 3$ (A1; " $-5 > x > 3$ " or " $3 < x < -5$ " is A0). [3]

2(a) $\theta = \frac{15}{12} = 1.25$ rad (B1). [1]

2(b) $\frac{1}{2}(12)^2(1.25) = 90 \text{ cm}^2$ (B1). [1]

2(c) Chord = $2 \times 12 \sin(0.625)$ (M1) = 14.0 cm (A1; 14.05). [Cosine rule acceptable.] [2]

3 $a + 7d = 3(a + d) \Rightarrow a = 2d$ (M1 A1). $a + 4d = 24 \Rightarrow 6d = 24 \Rightarrow d = 4, a = 8$ (A1). $S_{20} = 10(16 + 19 \times 4) = 920$ (A1). [4]

4(a) $2^6 + 6(2^5)x + 15(2^4)x^2$ (M1) = $64 + 192x + 240x^2$ (A1 A1). [3]

4(b) $(1 - 2x + x^2)(64 + 192x + 240x^2)$: x^2 coefficient = $240 - 2(192) + 64$ (M1) = -80 (A1). [2]

5(a) $gf(x) = \frac{8}{(2x+5)-1} = \frac{8}{2x+4}$ (M1) = $\frac{4}{x+2}$ (A1). [2]

5(b) $y = \frac{8}{x-1} \Rightarrow x-1 = \frac{8}{y} \Rightarrow x = 1 + \frac{8}{y}$ (M1). $g^{-1}(x) = 1 + \frac{8}{x}$ (A1). Domain $x > 0$ (B1). [3]

5(c) $g(x) > 0$ (B1). [1]

6(a) Midpoint $(2, 2)$ (B1). Gradient $AB = \frac{1-3}{5+1} = -\frac{1}{3}$, perpendicular gradient 3 (M1). $y - 2 = 3(x - 2) \Rightarrow y = 3x - 4$ (A1). [3]

6(b) $D(0, -4)$ (B1). Area: e.g. $\frac{1}{2} \times AB \times MD$ with $AB = \sqrt{40}$, $MD = \sqrt{4+36} = \sqrt{40}$ (M1) $\Rightarrow \frac{1}{2} \times 40 = 20$ (A1). [Shoelace: $\frac{1}{2}|(-1)(1+4) + 5(-4-3) + 0| = 20|$] [3]

7(a) $\frac{\sin \theta}{\cos \theta} \cdot \sin \theta + \cos \theta = \frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta}$ (M1 M1) = $\frac{1}{\cos \theta}$ (A1 AG). [3]

7(b) $\frac{1}{\cos \theta} = 4 \cos \theta \Rightarrow \cos^2 \theta = \frac{1}{4}$ (M1) $\Rightarrow \cos \theta = \pm \frac{1}{2}$ (A1). $\theta = 60^\circ, 300^\circ$ (A1), $120^\circ, 240^\circ$ (A1). [4]

8(a) $\frac{dy}{dx} = -12(2x-1)^{-2}$ (M1 A1). At $x = 1$: $y = 6$, gradient -12 : $y - 6 = -12(x - 1) \Rightarrow y = -12x + 18$ (A1). [3]

8(b) $\frac{dy}{dx} = -\frac{12}{(2x-1)^2} < 0$ for all $x > \frac{1}{2}$ since the square is positive (B1). [1]

8(c) At $x = 2$: $y = 2$, $\frac{dy}{dx} = -\frac{12}{9} = -\frac{4}{3}$ (B1). Normal gradient $\frac{3}{4}$ (M1). $y - 2 = \frac{3}{4}(x - 2) \Rightarrow 3x - 4y + 2 = 0$ (A1). [3]

9(a) $3 - 2 + k = 5 \Rightarrow k = 4$ (M1 A1). $y = x^3 - x^2 + 4x + c$ (M1); $(1, 2)$: $1 - 1 + 4 + c = 2 \Rightarrow c = -2$: $y = x^3 - x^2 + 4x - 2$ (A1). [4]

9(b) $\int (x^{1/2} + x^{-1/2}) dx = \frac{2}{3}x^{3/2} + 2x^{1/2}$ (M1 A1). $[\]_1^4 = (\frac{16}{3} + 4) - (\frac{2}{3} + 2)$ (M1) = $\frac{20}{3}$ (A1). [4]

10(a) $2 \times (\frac{3}{4})^3$ (M1) = $\frac{27}{32} = 0.844$ m (A1). [2]

10(b) Distance = $2 + 2(1.5 + 1.125 + 0.84375 + 0.6328)$, i.e. $2 + 2 \times \frac{1.5(1 - (3/4)^4)}{1 - 3/4}$ (M1 A1) =

$$2 + 8.20 = 10.2 \text{ m (A1). [3]}$$

$$\mathbf{10(c)} \quad \text{Total} = 2 + 2 \times \frac{1.5}{1 - 3/4} \text{ (M1 A1)} = 2 + 12 = 14 \text{ m (A1). [3]}$$

$$\mathbf{11(a)} \quad x^2 - x + 3 = 2x + c \Rightarrow x^2 - 3x + (3 - c) = 0 \text{ (M1). Discriminant } 9 - 4(3 - c) = 0 \text{ (M1 A1)} \\ \Rightarrow c = \frac{3}{4} \text{ (A1). [4]}$$

$$\mathbf{11(b)} \quad x = \frac{3}{2} \text{ (M1), } y = 2\left(\frac{3}{2}\right) + \frac{3}{4} = \frac{15}{4}: \text{ point } \left(\frac{3}{2}, \frac{15}{4}\right) \text{ (A1). [2]}$$

$$\mathbf{11(c)} \quad x^2 - 3x - 2 = 0 \Rightarrow x = \frac{3 \pm \sqrt{17}}{2} \text{ (M1). Difference in } x \text{ is } \sqrt{17}, \text{ difference in } y \text{ is } 2\sqrt{17} \text{ (A1).} \\ PQ = \sqrt{17 + 68} = \sqrt{85} \text{ (A1). [3]}$$

$$\mathbf{12(a)} \quad \text{Centre } (h, 0): (h - 2)^2 + 1 = (h - 6)^2 + 25 \text{ (M1)} \Rightarrow 8h = 56 \Rightarrow h = 7 \text{ (A1). } r^2 = (7 - 2)^2 + 1 = 26: (x - 7)^2 + y^2 = 26 \text{ (A1). [3]}$$

$$\mathbf{12(b)} \quad \text{Gradient of radius } CA = \frac{1-0}{2-7} = -\frac{1}{5}, \text{ tangent gradient } 5 \text{ (M1). } y - 1 = 5(x - 2) \Rightarrow y = 5x - 9 \text{ (A1). [2]}$$

$$\mathbf{12(c)} \quad \text{Midpoint } (4, 3), \text{ gradient } AB = 1, \text{ so perpendicular gradient } -1 \text{ (M1). } y - 3 = -(x - 4) \Rightarrow y = -x + 7 \text{ (A1). At } x = 7, y = 0, \text{ which is the centre } (7, 0) \checkmark \text{ (B1). [3]}$$

Total: 75